

PMIScore: An Unsupervised Approach to Quantify Dialogue Engagement

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Abstract

High dialogue engagement is a crucial indicator of an effective conversation. A reliable measure of engagement could help benchmark large language models, enhance the effectiveness of human-computer interactions, or improve personal communication skills. However, quantifying engagement is challenging, since it is subjective and lacks a “gold standard”. This paper proposes PMIScore, an efficient unsupervised approach to quantify dialogue engagement. It uses pointwise mutual information (PMI), which is the probability of generating a response conditioning on the conversation history. Thus, PMIScore offers a clear interpretation of engagement. As directly computing PMI is intractable due to the complexity of dialogues, PMIScore learned it through a dual form of divergence. The algorithm includes generating positive and negative dialogue pairs, extracting embeddings by large language models (LLMs), and training a small neural network using a mutual information loss function. We validated PMIScore on both synthetic and real-world datasets. Our results demonstrate the effectiveness of PMIScore in PMI estimation and the reasonableness of the PMI metric itself.

CCS Concepts

• **Theory of computation** → **Machine learning theory**; • **Mathematics of computing** → **Information theory**; • **Security and privacy** → *Economics of security and privacy*.

Keywords

Pointwise Mutual Information, Dialogue Engagement, Dialogue Benchmark

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1 Introduction

You sink into the couch, drained. “My presentation was a disaster.” Your partner barely glances over. “Yeah, traffic was awful today.”

Scenes like this happen every day—two people exchanging words, but not truly connecting. Sometimes one person answers a completely different question; other times, they respond with generic, safe phrases such as “I know how you feel” or “That’s interesting,” which keep the conversation going without real engagement.

Understanding and measuring engagement in natural language is more than a philosophical question. A reliable measure could help benchmark large language models, improve tools for business communication, track the quality of personal or professional interactions, or even analyze high-stakes scenarios like presidential debates [6, 25, 29]. However, quantifying engagement is subjective, often lacks a gold standard. Human annotations are noisy, expensive, and inconsistent. Directly asking a language model to score engagement is also unstable, hard to interpret, and can feel like a black-box solution [3].

In this paper, we focus on two research questions:

RQ1: Can we propose a metric to quantify the engagement of a dialogue?

RQ2: Do we have an efficient, automatic approach to calculate the metric without human annotations?

To address the questions, this paper proposes a pointwise mutual information (PMI) approach to measure engagement between conversational context and responses. PMI has a clear interpretation: it quantifies how much more (or less) likely a response is given the conversation context compared to its baseline probability.

$$\begin{aligned} \text{PMI}(\text{response}; \text{context}) &= \log \frac{\Pr[\text{response}, \text{context}]}{\Pr[\text{response}] \Pr[\text{context}]} \\ &= \log \frac{\Pr[\text{response}|\text{context}]}{\Pr[\text{response}]} \end{aligned} \quad (1)$$

In other words, if the response is highly relevant to the context, the PMI is positive; if the response is unrelated—or even counterintuitively discouraged by the context, as in the “traffic” example—the PMI can be negative. Generic, low-engagement responses, while valid in isolation, tend to be predictable regardless of the context, resulting in PMI scores close to zero.

Computing PMI requires estimating the joint distribution of conversation histories and responses—a task that becomes intractable in high-dimensional spaces. Unlike simple, low-dimensional signal systems, conversations are vast and complex, making naïve estimation impossible.

As large language models are based on the conditional probabilities to generate the texts, one might try using their log-probabilities directly. But this has limitations: those probabilities reflect the model’s general distribution over all contexts, not conversation-specific dynamics. They are also overly sensitive to surface-level context rather than meaningful exchange. Short texts are preferable by this method, even if they have the same semantic meaning.

There is a better path. Mutual information can be expressed as the KL-divergence between the joint distribution and the product of marginals: if context and response were independent, the two distributions would coincide. Pointwise mutual information (PMI) (Equation 1) corresponds to the log-ratio of these probabilities.

The dual form of KL-divergence provides a practical way to estimate this ratio. Instead of directly computing probabilities, we compare expectations of a function under the two distributions. If the distributions are identical, these expectations match; if not, there exists a function that maximizes their discrepancy. The optimal function corresponds exactly to the likelihood ratio, i.e., the PMI. In practice, expectations can be estimated via sampling:

- (1) **Joint samples:** context–response pairs from the same dialogue.
- (2) **Marginal samples:** contexts and responses drawn independently from different dialogues.

Viewed from another perspective, this setup resembles a classification problem. Positive samples are true context–response pairs, while negative samples are mismatched pairs. Training a discriminator to separate positives from negatives yields an approximation to PMI, quantifying how much mutual information a response shares with its conversational context. The key difference from standard classification is that our goal is not merely to minimize misclassification error, but to learn a score that directly corresponds to PMI. Achieving this requires an objective function derived from the dual form of KL-divergence.

Large language models further ease this training. Their pre-trained embeddings already encode rich semantic and contextual structure. By fine-tuning an LLM—or training a smaller classifier on top of its embeddings—we can specialize the model to this discriminative task. The fine-tuned model emphasizes conversational relevance over general fluency, aligning its internal representations with the objective of estimating mutual information. The detailed flow is shown in fig. 1.

Our contribution. We introduce a novel method for measuring dialogue engagement, grounded in information theory. Our approach is based on Pointwise Mutual Information (PMI), which quantifies how much new information the response contributes to the context. The method assigns a score of zero when the context and response are unrelated. For instance, if a response doesn’t offer any new or relevant insights—such as a generic, off-topic answer—it receives a score of zero, indicating no engagement. In contrast, when the response is highly relevant and aligned with the context, the score is positive, reflecting strong engagement. The score can turn negative if the response provides counterintuitive or misleading information that contradicts the context. This allows the metric to dynamically assess the quality of the response: it increases when additional context enhances the relevance of the response, but decreases when irrelevant or conflicting information detracts from it.

We also developed a practical, unsupervised method to calculate this measure, leveraging powerful language models (LLMs) and the duality format of mutual information. We proved that with some proper assumptions, our algorithm will converge to the PMI when the sample size goes to infinity.

Finally, we validated our approach using both synthetic datasets (created for testing) and real-world data. First, we showed that our method produced a more accurate estimation of PMI compared to previous methods. To establish a ground truth for PMI, we constructed synthetic datasets derived from various joint distributions. The results showed that our method has the lowest mean square error and the highest Pearson correlation with the true PMI. Furthermore, we demonstrated that PMIScore reliably measures dialogue engagement in the real-world dataset. We used the multilingual dialogue collections from DSTC-11 Track 4 to construct a training set. Then we evaluated the correlation between PMIScore and the human annotated relevance scores. Our results indicated that PMIScore exhibited a high correlation with the relevance score, thereby serving as a close and reliable metric for dialogue engagement.

2 Related work

Mutual Information and Dual Form. We use the f -divergence technique introduced by [14, 15]. Nguyen et al. [14] extends the correspondence between surrogate loss functions and f -divergences in the binary classification. [15] then develop a method to estimate the f -divergence of two probability distributions based on a non-asymptotic variational characterization. Using the same technique, [9] solve the learning, co-training, and forecast elicitation problem without verification simultaneously by maximizing the mutual information between agents. [2] also introduce MINE for estimating the mutual information in high-dimension data based on the dual form of KL-divergence. [33] propose a method that trains Bayesian models on embedded datasets to compute mutual information. In addition, [24] propose a family of scoring rules based on f -divergence to incentivize the truthful peer prediction.

We extend the method for estimating mutual information to the field of natural language processing using large pre-trained models. Additionally, we first propose using mutual information as a metric to measure dialogue engagement.

Benchmark Dialogue Systems. Many studies are dedicated to proposing metrics for measuring the capability of dialogue systems. Previous automatic benchmark includes BLEU ([17]), ROUGE ([12]), and BERTSCORE ([31]).

Nowadays, many researches explore the large language models (LLMs) to benchmark the dialogues. [13] introduces G-EVAL to evaluate summarization and dialogue response generation. [4] use GPTSCORE to test various evaluation aspects with zero-shot instruction. [3] propose a method to measure the engagement of dialogues by prompting the large language models.

Previous methods typically use prompt engineering to directly obtain a score from LLMs as a metric. Our approach utilizes an unsupervised fine-tuning method that uses mutual information as the metric, which offers better interpretability.

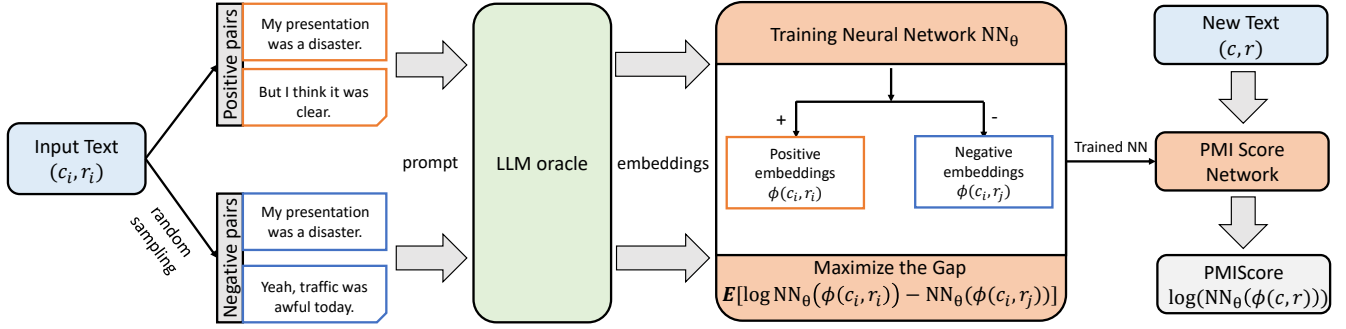


Figure 1: Flow of PMIScore.

Mutual Information in Dialogues. Numerous studies investigate how mutual information can be utilized to enhance the effectiveness of machine learning models, especially in the NLP system. [23] proposes a novel Discourse Mutual Information (DMI) loss function for pretraining dialog representation models, which capture conversational structure and response prediction uncertainty. [32] introduce a dialogue generation framework Dialog-MI which use the mutual information loss to generate more informative dialogue. [20] uses the conditional pointwise mutual information (C-PMI) to measure the turn-level interaction a dialogue systems. [26] train both natural language understanding and generation systems by maximizing the mutual information between two components. [28] also use a mutual information estimator to benchmark the judgment, especially for LLMs.

These researches often use mutual information as a training loss to improve the capability of a dialogue system, whereas we primarily use it as a metric to benchmark a dialogue system. Furthermore, previous methods typically used the upper bound of mutual information as an estimator, whereas we used its dual form, which allows for more accurate estimation of mutual information.

3 Preliminaries

In this section, we introduce some basic knowledge of mutual information that will be used in our relevance score and algorithm. We focus on the Shannon mutual information in the rest of our paper. We also consider a more general class of mutual information called f -mutual information. Similarly, we can also define the f -divergence and its duality. The details are deferred to the appendix.

DEFINITION 3.1 (MUTUAL INFORMATION). Given two random variables X, Y , the (Shannon) mutual information between X and Y is defined as

$$MI(X; Y) = \sum_{x, y} \Pr(x, y) \log \left(\frac{\Pr(x, y)}{\Pr(x) \Pr(y)} \right).$$

Mutual information (MI) quantifies the statistical dependence between two random variables X and Y . To analyze the association between specific outcomes (e.g., two texts), we use the *pointwise mutual information (PMI)*.

DEFINITION 3.2 (POINTWISE MUTUAL INFORMATION). For realizations $X = x$ and $Y = y$, the pointwise mutual information is defined

as

$$PMI(x; y) = \log \frac{\Pr(x, y)}{\Pr(x) \Pr(y)}.$$

In other words, $PMI(x; y)$ measures how much more likely the pair (x, y) is to occur together than if x and y were independent.

Moreover, mutual information is the expectation of PMI:

$$MI(X; Y) = \mathbb{E}_{(x, y) \sim \Pr(X, Y)} [PMI(x; y)].$$

Properties of PMI:

- **Symmetry** $PMI(x; y) = PMI(y; x)$.
- **Vanishing under independence:** $PMI(x; y) = 0$ if X and Y are independent.
- **Chain rule:** $PMI(x, y; z) = PMI(x; z) + PMI(y; z | x)$, with conditional $PMI(y; z | x) = \log \frac{\Pr(y, z | x)}{\Pr(y | x) \Pr(z | x)}$.
- **Monotonicity:** $PMI(x, y; z) \geq PMI(x; z)$ if $PMI(y; z | x) \geq 0$.
- **Sign:** $PMI(x; y) > 0$ if x and y co-occur more than expected; $PMI(x; y) < 0$ if less.

DEFINITION 3.3 (DUAL FORM OF MUTUAL INFORMATION). The KL-divergence between two distributions $\mathbf{p}$ and $\mathbf{q}$ is

$$d_{KL}(\mathbf{p} \| \mathbf{q}) = \mathbb{E}_{\mathbf{p}} \left[\log \frac{\mathbf{p}}{\mathbf{q}} \right].$$

It admits the dual form

$$d_{KL}(\mathbf{p} \| \mathbf{q}) = \sup_{D \in \mathcal{D}} \mathbb{E}_{\mathbf{p}} [\log D + 1] - \mathbb{E}_{\mathbf{q}} [D],$$

where $\mathcal{D}$ denotes the set of measurable functions $D : \mathcal{X} \rightarrow \mathbb{R}_{>0}$ and $\mathcal{X}$ denotes the underlying sample space of $\mathbf{p}$ and $\mathbf{q}$. Since mutual information is the KL-divergence between the joint distribution $\Pr(X, Y)$ and the product of marginals $\Pr(X) \Pr(Y)$, we obtain the dual form of mutual information:

$$MI(X; Y) = \sup_{D \in \mathcal{D}} (\mathbb{E}_{(x, y) \sim \Pr(X, Y)} [\log D(x, y) + 1] - \mathbb{E}_{(x, y) \sim \Pr(X) \Pr(Y)} [D(x, y)]). \quad (2)$$

Intuitively, this variational form interprets mutual information as the maximal advantage of a discriminator D in distinguishing dependent pairs from independent ones. At the optimum,

$$D^*(x, y) = \frac{\Pr(x, y)}{\Pr(x) \Pr(y)} = e^{PMI(x; y)},$$

which recovers the pointwise mutual information.

Why use the dual form? The direct definition of mutual information requires access to the joint probability $\Pr(x, y)$ and the marginals $\Pr(x), \Pr(y)$. In high-dimensional or continuous spaces, these distributions are intractable to compute or estimate accurately.

The dual form avoids this difficulty. Instead of explicitly estimating probability densities, it reframes mutual information as a variational optimization problem. Here, we only need *samples* from the joint distribution $\Pr(X, Y)$ (obtained from paired data) and from the product distribution $\Pr(X) \Pr(Y)$ (obtained by resampling or shuffling).

This variational perspective is the foundation of mutual information estimators such as *MINE* (Mutual Information Neural Estimation) and related approaches.

4 Framework of PMIScore

In this section, we introduce the **PMIScore** framework, which assigns a relevance score to each dialogue pair. A detailed description of the implementation is provided in the experiments section.

Conceptually, PMIScore estimates the pointwise mutual information (PMI) between a dialogue context and its response. Suppose we have n samples drawn from the joint distribution of contexts and responses, denoted as

$$\{(c_i, r_i)\}_{i=1}^n \sim \Pr(C, R).$$

PMIScore aims to quantify how much information the response r_i carries about its corresponding context c_i compared to responses drawn independently from the marginal distribution.

4.1 Data Preparation

There are n data points (c_i, r_i) in the corpus, where $c_i \in C$ is the context of a dialogue and $r_i \in R$ is the response text given c_i . For example, the context c_i usually contains the previous rounds of a dialogue and a question from another person, while r_i is the answer to that question.

The original dataset $D^+ = \{(c_i, r_i)\}_i$ is regarded as the *positive pairs*. and we construct the *negative pairs* $D^- = \{(c_i, r_j)\}_{i,j}$ by randomly sampling some response texts r_j from the dataset for each context c_i . As the response is meaningful but not to the context, we regard them as negatively relevance.

4.2 LLMs-Based Embeddings

Large Language Models (LLMs) are highly effective and powerful generative models. They are typically composed of two components: encoder blocks and decoder blocks. In traditional generation tasks, an LLM first receives an input. This input is processed by the encoder network to produce a vector (also called an embedding), which serves as a compressed representation of the input's information. Subsequently, this vector is passed through the decoder component, which then generates the response word by word in an autoregressive manner.

Given any pair of texts $d \in C \times R$, we use the encoder blocks of LLMs to extract the semantic meaning of the text. An LLM oracle is defined by $\phi : P \times C \times R \rightarrow \mathbb{R}^m$. It takes a task description (also called Prompt) p and a pair of data d as input and outputs a m -dimensional embedding e [8].

4.3 Training the Small Neural Network

Building on the PMIScore framework introduced above, we train a small neural network $\text{NN}_\theta : \mathbb{R}^m \rightarrow \mathbb{R}$ to compute the final relevance score for a given dialogue pair, where θ denotes the network parameters. The network is implemented as a multi-layer perceptron (MLP), making it significantly smaller and more efficient than full-scale LLMs.

Let D^+ denote the set of positive (context, response) pairs and D^- denote negative pairs. To align the network outputs with pointwise mutual information, we define a mutual information dual form-based loss function:

$$L_{\text{MI}}(\theta) = -\left(\mathbb{E}_{d \in D^+} \log \text{NN}_\theta(\phi(p^*, d)) - \mathbb{E}_{d \in D^-} \text{NN}_\theta(\phi(p^*, d))\right),$$

where p^* is a predetermined prompt and $\phi(\cdot)$ represents the input feature embedding. Details of the prompt and feature design are provided in the appendix.

Unlike traditional pairwise learning methods, this loss is asymmetric with respect to positive and negative pairs. This formulation directly corresponds to the dual-form of mutual information in Equation (2), with the only difference being that the constant term $+1$ from the original dual form is omitted. Since this constant does not affect the optimization, it can safely be dropped. As in the dual-form interpretation, this loss encourages the network to assign high scores to true context-response pairs while lowering the scores of mismatched pairs, effectively approximating the pointwise mutual information.

4.4 New Pairs Evaluation

After we trained the optimal parameters $\theta^* = \arg \min L_{\text{MI}}(\theta)$, we will use it to evaluate any pair of dialogues. Given the dialogue pair c, r , its PMIScore is given by:

$$\text{PMIScore}(c, r) = \log \text{NN}_{\theta^*}(\phi(p^*, (c, r))).$$

5 Theoretical Results

In this section, we prove that our algorithm will learn the PMI of each pair. Thus it is a proper metric for the relevance score. The main proof idea is borrowed from Belghazi et al. [2]. Due to limited space, we defer the proofs to Section B.

We first make two assumptions.

ASSUMPTION 5.1 (SEMANTIC EMBEDDING ASSUMPTION). *The LLM embedding $\phi(c, r)$ can extract the semantic meaning. That is, for any c_1, c_2 and r_1, r_2 with the same meaning, $\phi(c_1, r_1) = \phi(c_2, r_2)$.*

ASSUMPTION 5.2 (I.I.D. SAMPLES ASSUMPTION). *The data points $\{(c_i, r_i)\}_i$ are independent samples drawn from a joint distribution $\Pr(C = c, R = r)$.*

Notice that given the independent samples assumption, there are two underlying random variables C, R of the context c and response r separately. Now we state our main theorem which shows minimize our loss function L_{MI} is equivalent to estimate the mutual information of C, R . In addition, the minimizer corresponds to the PMI.

THEOREM 5.3. *Under the assumptions described above, as the number of samples $n \rightarrow \infty$, the PMIScore converges to the pointwise*

mutual information:

$$\text{PMIScore}(c, r) \longrightarrow \text{PMI}(c, r).$$

REMARK 5.4. *The PMIScore is symmetric with respect to context and response. A context-independent, “one-size-fits-all” response, which carries no information about the specific context, has $\Pr(Y = r \mid X = c) = \Pr(Y = r)$ and thus $\text{PMIScore}(c, r) = 0$. In contrast, responses that are highly context-specific yield positive scores.*

The chain rule of PMI implies that adding additional contextual information can further increase or decrease the pointwise mutual information. For example, if we extend the response from r to (r, z) , then

$$\text{PMI}(c; (r, z)) = \text{PMI}(c; r) + \text{PMI}(c; z \mid r),$$

The score increases when the additional context z provides positive information about the context, i.e., $\text{PMI}(c; z \mid r) > 0$, and decreases when z provides misleading or negatively correlated information, i.e., $\text{PMI}(c; z \mid r) < 0$. This highlights that PMIScore dynamically reflects the incremental contribution of new texts to the relevance of a context.

6 Synthetic Study

We run a controlled synthetic study to test whether PMIScore can recover the true pointwise mutual information (PMI) between a context and a response. Unlike the real dialogue datasets used later, this setting provides analytic ground truth for PMI.

6.1 Experimental Setup

Synthetic Distributions. We build three kinds of joint distributions $p(C, R)$ with different dependency patterns between context C and response R :

- (1) **Diagonal:** one-to-one aligned pairs ($C_i \leftrightarrow R_i$);
- (2) **Block:** pairs correlated within several topic blocks but independent across blocks;
- (3) **Independent:** C and R sampled independently from their marginals.

Each distribution yields an analytic PMI target. For every distribution we sample 5,000 pairs and split them into 60% train, 20% validation, and 20% test.

Negative Sampling. The training process uses both positive and negative pairs. **Positives** are the true (C, R) pairs sampled from $p(C, R)$. **Negatives** are formed by randomly mismatching C and R within the training positives, which mirrors real scenarios where we must derive negatives from observed data without access to $p(C)$ or $p(R)$. To reduce variance, we conduct 5 independent training runs, each using a distinct subset of 3 negatives per positive ($K=3$), yielding 15 negatives per positive in total. The validation and test sets contain only positives.

Embeddings and Scoring Network. Each context–response pair (c, r) is represented using sentence embeddings from several pretrained LLMs¹. A lightweight multilayer perceptron maps these embeddings to a scalar score approximating the true pointwise mutual information (PMI). More details about the training setting are shown in Section C.

¹Including Llama-3.2-3B, Phi-4-mini, Qwen3-0.6B/1.7B/4B/8B.

6.2 Compared Methods

We compare PMIScore with five baselines, yielding six methods in total: three mutual information estimators (**MINE**, **InfoNCE**, **KDE**), a dual-encoder variant of our method (**PMIScore-Pair**), and a state-of-the-art prompt-based evaluation method (**MEEP**).

MINE [7] is a variational mutual information estimator that optimizes the Donsker–Varadhan lower bound on the Kullback–Leibler divergence between the joint and product-of-marginals distributions. It relies on training a neural critic to maximize this bound, yielding flexible yet potentially unstable estimates when data are high-dimensional.

PMIScore-Pair is a dual-encoder variant of PMIScore that concatenates separately encoded context and response embeddings $[e_c; e_r]$ instead of using the unified prompt-based representation $\phi(c, r)$.

InfoNCE [16] formulates mutual information estimation as a contrastive prediction task: given a context c , the correct response r^+ must be distinguished from multiple negative samples $\{r^-\}$. The objective encourages higher similarity for true pairs and penalizes confusion with negatives via a temperature-scaled softmax. This formulation is efficient and widely used in self-supervised learning but is sensitive to negative sampling quality and temperature tuning.

KDE [18] is a classical non-parametric baseline that estimates densities via kernel density estimation. While conceptually simple, KDE struggles in high-dimensional embedding spaces, where density estimation becomes unreliable.

MEEP [3] is a state-of-the-art prompt-based dialogue evaluation method that directly prompts LLMs to generate numeric quality scores, representing the paradigm of using LLMs as evaluators without explicit MI estimation.

Together, these baselines span neural, contrastive, statistical, dual-encoder, and prompt-based paradigms.

Evaluation Metrics. Performance is evaluated primarily by **Mean Squared Error (MSE)** between predicted and ground-truth PMI, complemented by **Pearson Correlation** to assess the linear alignment.

6.3 Results and Analysis

As shown in Figure 2, PMIScore and its dual-encoder variant consistently achieve lower MSE and higher correlation than other baselines across all dependency structures (*Diagonal*, *Block*, and *Independent*). This indicates that PMIScore not only recovers the correct ordering of mutual information strength but also provides more accurate quantitative estimates. Figure 3 further illustrates the estimation behavior on the *Block* dataset: predictions from PMIScore align closely with the identity line, while MINE and InfoNCE exhibit systematic bias or slope distortion, and KDE shows wide dispersion due to high-dimensional estimation noise.

Notably, **PMIScore-Pair** achieves comparable or even slightly better performance than the prompt-based PMIScore on synthetic data. This is because the synthetic distributions are relatively simple and low-dimensional: context–response dependencies can be captured by concatenating separate embeddings without requiring deep self-attention within the Transformer. In contrast, **MEEP** performs poorly across all synthetic settings, as it relies on the

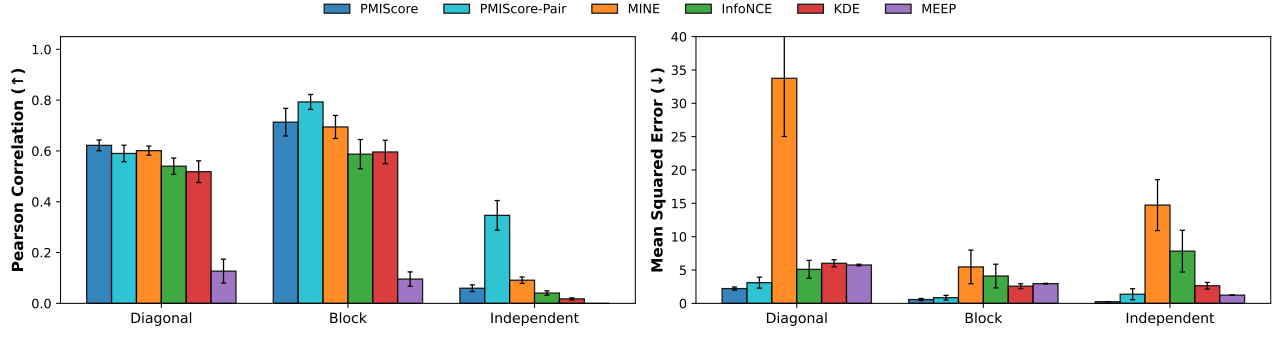


Figure 2: PMIScore vs. baselines on synthetic distributions. The figure shows the Pearson correlation (left) and mean squared error (right) of six methods (PMIScore, PMIScore-Pair, MINE, InfoNCE, KDE, and MEEP) on three synthetic dependency structures (*Diagonal*, *Block*, and *Independent*). Higher Pearson values indicate better alignment with the true PMI and lower MSE means better approximation. PMIScore achieves consistently high correlation and low MSE across almost all settings.

LLM’s pretrained knowledge to generate quality scores directly from prompts. Since the synthetic data is artificially constructed with predefined prototypes and paraphrasing, it falls outside the natural dialogue distribution that LLMs were trained on, rendering pure prompt-based evaluation ineffective.

Overall, the synthetic results validate that PMIScore aligns with the theoretical definition of PMI while remaining robust to variations in dependency structure and embedding model.

7 Empirical Study

This section evaluates whether PMIScore learned in real multi-turn dialogues can (i) rank the true response above hard negatives and (ii) align with human-perceived relevance. Unlike the synthetic setting—where ground-truth PMI is analytically available—here we rely on task performance and human judgments as proxies. The pipeline mirrors the synthetic study: we embed (c, r) pairs with LLMs, train a small scorer with the dual-form MI objective (PMIScore), and compare against **PMIScore-Pair**, **MINE**, **InfoNCE**, **KDE**, and **MEEP**.

7.1 Experimental Setup

Datasets. We evaluate on the multilingual dialogue collections from DSTC-11 Track 4 (dstc_en, dstc_zh; [22]). Some representative sources include: English datasets such as DailyDialog [11], PersonaChat/ConvAI2 [30], EmpatheticDialogues [19], and TopicalChat [5]; Chinese datasets such as LCCC [27] and KdConv [34]. Each example uses a multi-turn context c (dialogue history) paired with its response r^+ .

Data split. For each language, we randomly sample 3,000/1,000/1,000 context–response pairs for *train/validation/test* from the DSTC-11 Track 4 training pool. Model selection is given by *validation Area Under the Curve (AUC)*; all retrieval results are reported on the held-out test split.

Negative construction. For each positive pair (c, r^+) , negatives are sampled according to a Bernoulli-like distribution:

- **In-dialogue negative** (prob = 0.1): A different turn from the *same* dialogue that is not the gold continuation, serving as a hard distractor.
- **Random negative** (prob = 0.9): A response sampled from the corpus outside the current dialogue to ensure diversity.

This stochastic mixture balances contextual difficulty with global contrast while avoiding label leakage. To reduce variance, we conduct 5 independent training runs, each using a distinct subset of 3 negatives per positive ($K=3$), yielding 15 negatives per positive in total.

Human annotations. For correlation analysis we use the human-annotated dev packs included in Track 4. We take the averaged relevance field `annot_relevant_mean` as the human score and evaluate per LLM on $n=375$ English and $n=1,589$ Chinese pairs, respectively.

LLMs and embeddings. We probe scaling with compact encoders: Qwen3 (0.6B/1.7B/4B/8B), Llama-3.2-3B-Instruct, and Phi-4-mini. For each (c, r) , sentence embeddings are fed to a lightweight MLP scorer. Architecture and preprocessing mirror the synthetic study for an apples-to-apples comparison.

Training protocol. The only difference from the synthetic setup is the absence of analytic PMI. All other ingredients are identical: the negative recipe, batching, optimizer, and early stopping by validation AUC.

7.2 Metrics

(1) **ROC-AUC (response ranking).** Given a context c , we score the gold response r^+ and its negatives $\{r^-\}$. ROC-AUC is the probability that a randomly chosen positive is ranked above a randomly chosen negative; it is computed from the score distributions without choosing a threshold. We report overall AUC and show averages across LLMs in Fig. 4.

(2) **Spearman correlation (human alignment).** To evaluate PMIScore’s alignment to human labels, we compute Spearman’s ρ between model scores and human annotated relevance, and we show the average across LLMs.

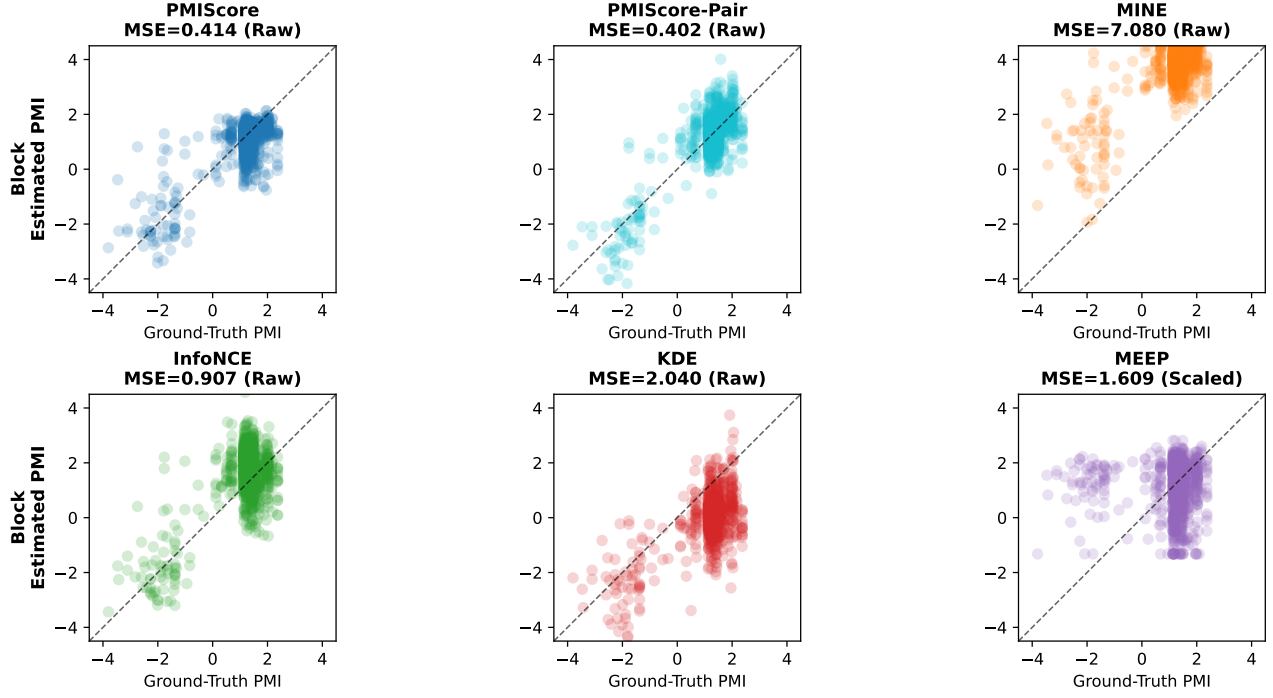


Figure 3: Consistency of PMI estimation for different methods. Each panel compares the estimated versus ground-truth pointwise mutual information (PMI) for six estimation methods—PMIScore, PMIScore-Pair, MINE, InfoNCE, KDE, and MEEP—on the *Block* synthetic dataset with the Qwen3-4B embeddings. To illustrate the magnitude of estimation error, 1,000 representative samples are plotted in each figure. The x-axis is the ground-truth of PMI and the y-axis is the estimated value. The dashed line means that the estimated PMI perfectly matches the ground truth. While PMIScore and PMIScore-Pair align closely with the identity line, indicating minimal estimation bias, alternative estimators display noticeable deviations or slope distortions. Similar patterns are also observed across other embedding models and synthetic datasets.

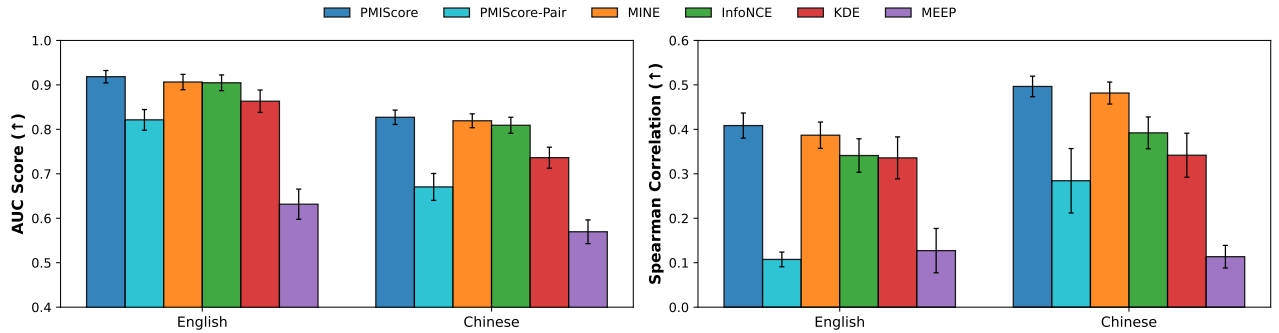


Figure 4: PMIScore vs. baselines on empirical dialogue datasets. Bars show mean across different LLMs for English (dstc_en) and Chinese (dstc_zh). *Left*: ROC-AUC on the *test* split for the response-ranking task. *Right*: Spearman correlation with human relevance annotations. Six methods are compared: PMIScore, PMIScore-Pair, MINE, InfoNCE, KDE, and MEEP. PMIScore achieves the highest AUC on both English and Chinese, and the best human correlation on both languages.

7.3 Results

Aggregated AUC. As summarized in Fig. 4 (left), **PMIScore** achieves the highest AUC on English and Chinese, closely followed by **MINE** and **InfoNCE**. All three prompt-based neural estimators

substantially outperform the non-parametric **KDE** and the purely prompt-based **MEEP**. Notably, **PMIScore-Pair** (dual-encoder) lags

behind its prompt-based counterpart by a clear margin on both languages. This contrasts with the synthetic setting where PMIScore-Pair performed comparably, suggesting that real dialogue data requires deeper context–response interaction within the Transformer to capture nuanced semantic dependencies.

Human alignment. Fig. 4 (right) shows that **PMIScore** attains the best Spearman correlation with human relevance judgments on both English and Chinese. **MINE** follows as a strong second, while **InfoNCE** and **KDE** trail further behind. **PMIScore-Pair** exhibits particularly weak human alignment, reinforcing that simple embedding concatenation fails to capture the pragmatic cues humans rely on. **MEEP**, despite leveraging LLM prompting, also underperforms, likely because its zero-shot scoring lacks task-specific calibration.

Takeaways. (i) **Ranking.** PMIScore leads or matches top baselines on AUC across both languages, clearly outperforming KDE and MEEP. (ii) **Human alignment.** PMIScore achieves the best correlation on both languages, demonstrating that PMI-based scoring aligns well with human perception of relevance. (iii) **Prompt vs. Pair.** The prompt-based representation consistently outperforms the dual-encoder variant on real dialogues. Unlike the simpler synthetic distributions, real conversational data benefits from deep cross-attention that fuses context and response before scoring.

8 Conclusion

In this paper, we propose the PMIScore to measure the engagement of a dialogue. It generates positive and negative samples from the dialogue corpus, extracts the semantic embeddings by LLMs and trains a neural network by mutual information based loss functions. We justify its theoretical interpretation and validate its performance on both synthetic and real-world datasets.

There are several future directions to improve our work. First, we will explore the application of PMIScore to enhance dialogue generation agents. This can be achieved by integrating it as a reward function within a reinforcement learning framework or by employing it as a fine-tuning objective for large language models (LLMs).

Second, as real-world conversations often require domain-specific adaptations in real time, such as interactions with strangers or subject matter experts, we can extend our framework to an online learning setting. This will enable the system to quickly adapt to new conversational styles and dynamics.

Finally, since our theoretical results rely on the semantic assumptions of LLMs, a critical area for future research is to investigate methods for obtaining more robust embeddings. This is essential for mitigating potential errors in PMIScore and ensuring the reliability of our model.

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A Appendix

A.1 f -Mutual Information

DEFINITION A.1 (f -MUTUAL INFORMATION [10]). *Given two random variables X, Y , the f -mutual information between X and Y is defined as*

$$MI^f(X; Y) = \sum_{x, y} \Pr(x) \Pr(y) f\left(\frac{\Pr(x, y)}{\Pr(x) \Pr(y)}\right),$$

where f is a convex function and $f(1) = 0$.

By picking $f(t) = t \log t$, we obtain the classic Shannon mutual information,

$$MI(X; Y) = \sum_{x, y} \Pr(x, y) \log\left(\frac{\Pr(x, y)}{\Pr(x) \Pr(y)}\right).$$

f -Mutual Information has some desired properties.

LEMMA A.2 (PROPERTIES OF f -MUTUAL INFORMATION [10]). *f -Mutual information satisfies*

Symmetry: $MI^f(X; Y) = MI^f(Y; X)$;

Non-negativity: $MI^f(X; Y)$ is always non-negative and is 0 if X is independent of Y ;

Information Monotonicity: $MI^f(T(X); Y) \leq MI^f(X; Y)$ where $T(\cdot) \in \mathbb{R}^{|\mathbb{X}| \times |\mathbb{X}|}$ is a possibly random operator on X whose randomness is independent of Y .

Now we introduce f -divergence, Fenchel's duality and their relation to f -mutual information.

DEFINITION A.3 (FENCHEL DUALITY [21]). *Given any function $f : \mathbb{R} \mapsto \mathbb{R}$, we define its convex conjugate f^* as a function that also maps $\mathbb{R}$ to $\mathbb{R}$ such that*

$$f^*(x) = \sup_t tx - f(t).$$

DEFINITION A.4 (f -DIVERGENCE [1]). *f -Divergence $d_f : \Delta_\Sigma \times \Delta_\Sigma \mapsto \mathbb{R}$ is a non-symmetric measure of the difference between distribution $\mathbf{p} \in \Delta_\Sigma$ and distribution $\mathbf{q} \in \Delta_\Sigma$ and is defined to be*

$$d_f(\mathbf{p}; \mathbf{q}) = \sum_{\sigma \in \Sigma} \mathbf{q}(\sigma) f\left(\frac{\mathbf{p}(\sigma)}{\mathbf{q}(\sigma)}\right),$$

where $f : \mathbb{R} \mapsto \mathbb{R}$ is a convex function and $f(1) = 0$.

As an example, by picking $f(t) = t \log(t)$, we obtain the well-known KL-divergence $d_{KL}(\mathbf{p}, \mathbf{q}) = \sum_{\sigma} \mathbf{p}(\sigma) \log \frac{\mathbf{p}(\sigma)}{\mathbf{q}(\sigma)}$.

The f -mutual information can be understood as the f -divergence between the joint distribution $\Pr(X = x, Y = y)$, denoted as XY , and the product of the marginal distribution $\Pr(X = x) \Pr(Y = y)$, denoted as $X \otimes Y$.

LEMMA A.5. *We have the equation*

$$MI^f(X; Y) = d_f(XY; X \otimes Y).$$

Now we introduce the dual form of f -divergence.

LEMMA A.6 (DUAL FORM OF f -DIVERGENCE [15]).

$$d_f(\mathbf{p}; \mathbf{q}) = \sup_{D \in \mathcal{D}} \mathbb{E}_{\mathbf{p}} \partial f(D) - \mathbb{E}_{\mathbf{q}} f^*(\partial f(D))$$

where $\mathcal{D}$ is a set of functions that maps Σ to $\mathbb{R}$ and the best D^* satisfies $D^*(\sigma) = \frac{\mathbf{p}(\sigma)}{\mathbf{q}(\sigma)}$. f^* is the Fenchel Duality of f .

B Omitted Proofs

PROOF OF THEOREM 5.3. We prove the theorem by two lemmas from [2].

LEMMA B.1 (UNIVERSAL APPROXIMATION [2]). *For any $\epsilon > 0$, there exists a neural network parametered by $\theta \in \Theta$ to approximate the optimal mutual information. Suppose $\sup_{\theta \in \Theta} -L_{MI}(\theta) = MI(\Theta)$, then*

$$|MI(\Theta) - MI(C, R)| < \epsilon, a.e.$$

LEMMA B.2 (ESTIMATION [2]). *Suppose $\sup_{\theta \in \Theta} -L_{MLD(n)}(\theta) = MI_n(\Theta)$. For any $\epsilon > 0$, there exists an integer N such that for any $n > N$,*

$$|MI_n(\Theta) - MI(\Theta)| < \epsilon, a.e.,$$

where $-L_{MLD(n)}(\theta)$ is the loss function defined in the dataset $D^{(n)}$ with size n .

Combining these lemmas and using the triangle inequality, we have

$$\mathbb{E}[\text{PMIScore}(c, r)] \rightarrow MI(C, R).$$

Since $\mathbb{E}[\text{PMIScore}(c, r)] \leq MI(C, R)$ and the equation holds only if $\text{PMIScore}(c, r) = \text{PMI}(c, r)$. Furthermore, we have

$$\text{PMIScore}(c, r) \rightarrow \text{PMI}(c, r).$$

□

C Synthetic Study: Additional Details and Results

C.1 Additional Details

Datasets. We study three synthetic corpora that impose different dependency structures over context–response pairs: *Diagonal*, *Block*, and *Independent*. For each dataset we predefine **20** context prototypes $\{\tilde{c}_i\}_{i=1}^{20}$ and **20** response prototypes $\{\tilde{r}_j\}_{j=1}^{20}$ (see code’s prototypes for concrete strings). A dataset-specific joint distribution over indices $(i, j) \in \{1, \dots, 20\}^2$ governs sampling:

- **Diagonal:** mass concentrated on $i = j$ (approximately one-to-one alignment);
- **Independent:** product distribution $\pi(i, j) = \pi_X(i)\pi_Y(j)$ (no dependency);
- **Block:** block-structured coupling between topical groups of contexts and generic supportive responses.

Each sampled prototype pair $(\tilde{c}_i, \tilde{r}_j)$ is then **paraphrased by an LLM** to produce (c, r) ; all embeddings and training use the paraphrased (c, r) (not the raw prototypes). This enlarges the support, better matches real text diversity, and reduces overfitting. Negatives are created by random mismatching within the same dataset; each positive is paired with **15** negatives.

Encoders and Representations. Each pair (c, r) is encoded via a single prompt that forms one input sequence for a frozen LLM encoder. We use the following prompt verbatim:

```
You are an assistant skilled at evaluating the
relevance of a response to a given context.
Task: Evaluate the relevance of the following
response to the context.
Context: {context_text}
Response: {response_text}
Result:
```

The encoder processes this prompt (with $\{\text{context_text}\}$ and $\{\text{response_text}\}$ filled by c and r) and outputs a single representation vector $\phi(c, r)$ for the *pair*. All embeddings are produced locally (vLLM). Encoders: Llama-3.2-3B-Instruct, Phi-4-mini-instruct, Qwen3- $\{0.6\text{B}, 1.7\text{B}, 4\text{B}, 8\text{B}\}$.

Scoring Network. On top of the frozen encoder, we train a compact MLP head that maps $\phi(c, r)$ to a score $\text{NN}_\theta(\phi(c, r))$:

$$\begin{aligned} &\text{Linear}(d, 256) \rightarrow \text{PReLU} \\ &\rightarrow \text{Linear}(256, 128) \rightarrow \text{PReLU} \\ &\rightarrow \text{Linear}(128, 1) \rightarrow \text{softcap}_{20} \end{aligned}$$

where $\text{softcap}_{20}(x) = 20 \tanh(x/20)$.

Training Setup. We use AdamW. A unified rule sets the learning rate for *all* encoders:

$$\text{lr} = 1 \times 10^{-3} \times \frac{1024}{d},$$

with d the encoder embedding dimension. We train for **100** epochs with positive-batch size **256** and **3** negatives per positive ($K=3$).

Objective Functions. Let $\mathcal{P}_+$ and $\mathcal{P}_-$ be the positive (true) and negative (mismatched) pair distributions. The model outputs $\text{NN}_\theta(\phi(c, r))$

on the prompt-based pair embedding $\phi(c, r)$. Consistent with the implementation:

- **PMIScore**

$$\mathcal{L}_{\text{PMI}} = -\left(\mathbb{E}_{(c,r) \sim \mathcal{P}_+} [\text{NN}_\theta(\phi(c, r))] - \mathbb{E}_{(c,r^-) \sim \mathcal{P}_-} [\exp(\text{NN}_\theta(\phi(c, r^-)))]\right).$$

- **InfoNCE**

$$\mathcal{L}_{\text{NCE}} = -\mathbb{E} \left[\log \frac{\exp(\text{NN}_\theta(\phi(c, r)))}{\exp(\text{NN}_\theta(\phi(c, r))) + \sum_{k=1}^K \exp(\text{NN}_\theta(\phi(c, r_k^-)))} \right].$$

- **MINE**

$$\mathcal{L}_{\text{MINE}} = -\left(\mathbb{E}_{(c,r) \sim \mathcal{P}_+} [\text{NN}_\theta(\phi(c, r))] - \log \mathbb{E}_{(c,r^-) \sim \mathcal{P}_-} [\exp(\text{NN}_\theta(\phi(c, r^-)))]\right).$$

- **KDE** is a non-parametric baseline:

$$\text{KDE}(\phi(c, r)) = \log \hat{p}_{\text{pos}}(\phi(c, r)) - \log \hat{p}_{\text{neg}}(\phi(c, r))$$

where $\hat{p}_{\text{pos}}$ and $\hat{p}_{\text{neg}}$ are Gaussian kernel density estimators fitted on positive and negative pair embeddings respectively, with Scott’s rule for bandwidth selection.

C.2 Additional Results

Metrics. We report Pearson’s ρ (higher is better) and mean squared error (MSE; lower is better) between predicted scores and ground-truth synthetic PMI.

Results. Table 1 shows results for the six methods by dataset and encoder. Within each model group, the best ρ (max) and MSE (min) are **boldfaced**. PMIScore is generally strongest, especially on *Diagonal* and *Block*.

D Empirical Study: Additional Details and Results

D.1 Additional Details

Datasets and splits. We evaluate on the multilingual dialogue collections from DSTC-11 Track 4 (English `dstc_en`, Chinese `dstc_zh`; see §7). Per language we draw 3,000/1,000/1,000 context–response pairs for *train/validation/test* from the Track 4 training pool, fixing a global seed (42) for reproducibility. Model selection uses *validation AUC*; all retrieval numbers reported in the main text are on the held-out *test* split. Human correlation is computed only on the official human-annotated dev packs (English $n=375$, Chinese $n=1,589$).

Preprocessing of contexts and responses. We keep multi-turn histories. For each dialogue with turns $\{u_1, \dots, u_i, \dots\}$, a positive pair is formed as $\text{context} = u_1 \oplus \dots \oplus u_{i-1}$ (newline-joined, no speaker tags) and $\text{response} = u_i$ (stripped of any speaker prefixes).² This yields realistic, variable-length contexts and naturally hard in-dialogue negatives.

²The loader normalizes common JSON/CSV variants (context/history/dialog for contexts, response/reference/answer for responses), converts lists to text, removes empty lines, and trims whitespace.

Dataset	Model	PMIScore		PMIScore-Pair		MINE		InfoNCE		KDE		MEEP	
		ρ	MSE	ρ	MSE	ρ	MSE	ρ	MSE	ρ	MSE	ρ	MSE
Block	Phi-4-mini-instruct	0.802 (0.017)	0.350 (0.070)	0.850 (0.006)	0.325 (0.072)	0.754 (0.023)	2.520 (3.499)	0.662 (0.042)	1.686 (1.042)	0.702 (0.016)	1.923 (0.036)	0.124 (0.021)	2.902 (0.038)
	Qwen3-0.6B	0.470 (0.019)	1.287 (0.424)	0.652 (0.027)	2.575 (0.912)	0.533 (0.012)	14.253 (16.473)	0.370 (0.016)	12.089 (13.000)	0.390 (0.009)	4.268 (0.019)	-0.002 (0.043)	3.119 (0.079)
	Qwen3-1.7B	0.649 (0.038)	0.692 (0.331)	0.783 (0.009)	0.699 (0.163)	0.578 (0.019)	12.457 (8.151)	0.464 (0.027)	5.630 (3.639)	0.550 (0.021)	2.677 (0.028)	0.037 (0.042)	3.058 (0.069)
	Qwen3-4B	0.777 (0.017)	0.371 (0.064)	0.821 (0.008)	0.443 (0.040)	0.742 (0.029)	2.080 (2.841)	0.639 (0.022)	0.850 (0.053)	0.638 (0.009)	2.059 (0.061)	0.096 (0.045)	2.964 (0.075)
	Qwen3-8B	0.811 (0.010)	0.337 (0.047)	0.823 (0.010)	0.406 (0.040)	0.808 (0.019)	0.398 (0.075)	0.750 (0.011)	0.702 (0.262)	0.632 (0.024)	2.533 (0.104)	0.191 (0.035)	2.789 (0.077)
	Llama-3.2-3B-Instruct	0.771 (0.017)	0.404 (0.106)	0.828 (0.007)	0.632 (0.059)	0.752 (0.014)	1.055 (1.130)	0.639 (0.061)	3.618 (3.860)	0.664 (0.017)	1.986 (0.065)	0.128 (0.027)	2.893 (0.050)
	Llama-3.2-3B-Instruct	0.771 (0.017)	0.404 (0.106)	0.828 (0.007)	0.632 (0.059)	0.752 (0.014)	1.055 (1.130)	0.639 (0.061)	3.618 (3.860)	0.664 (0.017)	1.986 (0.065)	0.128 (0.027)	2.893 (0.050)
Diagonal	Phi-4-mini-instruct	0.664 (0.009)	1.728 (0.251)	0.660 (0.012)	1.523 (0.183)	0.618 (0.010)	44.496 (11.139)	0.561 (0.044)	2.786 (1.086)	0.609 (0.021)	5.167 (0.095)	0.120 (0.034)	5.781 (0.083)
	Qwen3-0.6B	0.525 (0.012)	3.210 (0.934)	0.438 (0.237)	7.045 (2.866)	0.518 (0.025)	58.645 (91.388)	0.387 (0.026)	8.500 (6.521)	0.381 (0.010)	8.336 (0.041)	0.021 (0.022)	6.021 (0.054)
	Qwen3-1.7B	0.608 (0.008)	2.545 (0.762)	0.571 (0.018)	2.924 (0.139)	0.589 (0.021)	52.908 (11.390)	0.545 (0.030)	9.884 (8.883)	0.392 (0.009)	6.756 (0.086)	0.048 (0.024)	5.926 (0.082)
	Qwen3-4B	0.629 (0.007)	1.942 (0.257)	0.619 (0.027)	2.106 (0.353)	0.611 (0.014)	26.183 (9.451)	0.549 (0.015)	4.612 (1.841)	0.543 (0.007)	5.642 (0.080)	0.050 (0.035)	5.948 (0.161)
	Qwen3-8B	0.659 (0.008)	1.769 (0.207)	0.626 (0.012)	1.982 (0.296)	0.637 (0.007)	7.802 (3.534)	0.598 (0.036)	2.397 (1.459)	0.594 (0.010)	5.115 (0.114)	0.321 (0.029)	5.231 (0.067)
	Llama-3.2-3B-Instruct	0.647 (0.017)	2.093 (0.535)	0.625 (0.027)	3.028 (0.382)	0.634 (0.017)	12.449 (2.647)	0.598 (0.028)	2.400 (0.456)	0.590 (0.004)	5.028 (0.129)	0.200 (0.021)	5.578 (0.053)
	Llama-3.2-3B-Instruct	0.647 (0.017)	2.093 (0.535)	0.625 (0.027)	3.028 (0.382)	0.634 (0.017)	12.449 (2.647)	0.598 (0.028)	2.400 (0.456)	0.590 (0.004)	5.028 (0.129)	0.200 (0.021)	5.578 (0.053)
Independent	Phi-4-mini-instruct	0.110 (0.009)	0.255 (0.035)	0.082 (0.091)	0.229 (0.012)	0.098 (0.014)	27.308 (52.609)	0.079 (0.026)	5.817 (6.019)	0.014 (0.011)	3.334 (0.130)	-0.009 (0.029)	1.227 (0.026)
	Qwen3-0.6B	0.018 (0.034)	0.228 (0.009)	0.310 (0.038)	5.446 (2.557)	0.061 (0.013)	11.448 (9.988)	0.034 (0.017)	23.007 (40.962)	0.021 (0.019)	1.479 (0.007)	-0.008 (0.013)	1.231 (0.011)
	Qwen3-1.7B	0.076 (0.027)	0.252 (0.013)	0.367 (0.030)	0.591 (0.088)	0.128 (0.032)	21.328 (16.693)	0.045 (0.033)	7.844 (11.021)	0.001 (0.021)	1.959 (0.032)	-0.006 (0.029)	1.223 (0.025)
	Qwen3-4B	0.046 (0.018)	0.256 (0.006)	0.425 (0.011)	0.554 (0.127)	0.051 (0.017)	19.034 (25.263)	0.033 (0.016)	3.188 (3.290)	0.029 (0.018)	1.699 (0.018)	-0.025 (0.016)	1.242 (0.018)
	Qwen3-8B	0.068 (0.027)	0.253 (0.008)	0.405 (0.025)	0.398 (0.075)	0.089 (0.030)	5.926 (7.212)	0.015 (0.023)	3.545 (1.001)	0.020 (0.021)	4.732 (0.186)	-0.027 (0.025)	1.246 (0.025)
	Llama-3.2-3B-Instruct	0.039 (0.023)	0.252 (0.010)	0.490 (0.032)	0.987 (0.087)	0.119 (0.024)	3.357 (1.649)	0.036 (0.015)	3.571 (4.536)	0.022 (0.016)	2.617 (0.099)	0.013 (0.022)	1.209 (0.020)
	Llama-3.2-3B-Instruct	0.039 (0.023)	0.252 (0.010)	0.490 (0.032)	0.987 (0.087)	0.119 (0.024)	3.357 (1.649)	0.036 (0.015)	3.571 (4.536)	0.022 (0.016)	2.617 (0.099)	0.013 (0.022)	1.209 (0.020)

Table 1: Performance on Synthetic datasets. Best Pearson ρ (higher) and MSE (lower) per model are bolded.

Prompted pair encoding (frozen encoders). We encode each (c, r) with a single *pair-level* prompt to a frozen LLM encoder, producing one vector $\phi(c, r)$ per pair:

```
You are an assistant skilled at evaluating the
relevance of a response to a given context.
Task: Evaluate the relevance of the following
response to the context.
Context: {context_text}
Response: {response_text}
Result:
```

Unless otherwise noted, we use the *English* template for *both* languages for consistency.³ We use vLLM (task=embed) to process the embeddings.

LLMs (frozen). Encoders: Qwen3-{0.6B, 1.7B, 4B, 8B}, Llama-3.2-3B-Instruct,

³A Chinese template is available in code but disabled by default.

microsoft/Phi-4-mini-instruct. We only train the small scorer head; encoders remain frozen throughout.

Scorer head (shared across methods). A lightweight MLP maps $\phi(c, r) \in \mathbb{R}^d$ to $NN_\theta(c, r) \in \mathbb{R}$, the structure is the same as the MLP in synthetic study.

Objectives. Given positives $\mathcal{P}_+$ and (constructed) negatives $\mathcal{P}_-$, losses follow App. C: PMIScore (dual-form), InfoNCE (batch-wise contrastive), MINE (DV bound). KDE is non-parametric: we fit StandardScaler $\rightarrow$ optional PCA (128 dims) $\rightarrow$ Gaussian KDE for p_+ and p_- with an automatic bandwidth heuristic; the score is $\log p_+ - \log p_-$.

Optimization and early stopping. We use AdamW with learning rate $\text{lr} = 10^{-3} \times (1024/d)$ where d is the encoder embedding size. We train for up to 100 epochs, with positive batch size 256 and NEG_RATIO= 3 (so each *positive* batch is matched with a *negative* batch of size $3 \times$ for InfoNCE). We select checkpoints by *validation*

Dataset	Model	PMIScore		PMIScore-Pair		MINE		InfoNCE		KDE		MEEP	
		AUC	ρ	AUC	ρ	AUC	ρ	AUC	ρ	AUC	ρ	AUC	ρ
En	Phi-4-mini-instruct	0.922	0.407	0.841	0.102	0.910	0.379	0.919	0.436	0.881	0.482	0.673	0.335
		(0.002)	(0.096)	(0.008)	(0.118)	(0.005)	(0.107)	(0.002)	(0.075)	(0.002)	(0.052)	(0.010)	(0.085)
	Qwen3-0.6B	0.858	0.286	0.712	0.073	0.829	0.258	0.830	0.175	0.781	0.214	0.531	0.129
		(0.006)	(0.075)	(0.137)	(0.170)	(0.026)	(0.081)	(0.009)	(0.069)	(0.005)	(0.092)	(0.013)	(0.127)
	Qwen3-1.7B	0.903	0.433	0.810	0.061	0.892	0.406	0.885	0.321	0.793	0.180	0.593	-0.018
		(0.004)	(0.055)	(0.071)	(0.064)	(0.003)	(0.092)	(0.006)	(0.078)	(0.004)	(0.060)	(0.018)	(0.082)
	Qwen3-4B	0.941	0.475	0.865	0.098	0.933	0.445	0.936	0.351	0.914	0.381	0.670	0.038
Zh		(0.002)	(0.028)	(0.006)	(0.069)	(0.003)	(0.024)	(0.006)	(0.090)	(0.003)	(0.076)	(0.016)	(0.110)
	Qwen3-8B	0.952	0.467	0.853	0.151	0.947	0.462	0.952	0.414	0.926	0.365	0.798	0.173
		(0.003)	(0.032)	(0.009)	(0.093)	(0.003)	(0.023)	(0.003)	(0.045)	(0.004)	(0.074)	(0.008)	(0.153)
	Llama-3.2-3B-Instruct	0.934	0.384	0.848	0.160	0.926	0.371	0.907	0.349	0.884	0.393	0.653	0.105
		(0.003)	(0.069)	(0.004)	(0.040)	(0.005)	(0.058)	(0.009)	(0.104)	(0.002)	(0.095)	(0.009)	(0.090)
	Phi-4-mini-instruct	0.824	0.475	0.692	0.349	0.812	0.453	0.799	0.405	0.741	0.389	0.575	0.128
		(0.003)	(0.047)	(0.007)	(0.033)	(0.006)	(0.045)	(0.006)	(0.037)	(0.002)	(0.037)	(0.011)	(0.060)
Zh	Qwen3-0.6B	0.774	0.430	0.521	-0.076	0.776	0.404	0.761	0.388	0.682	0.170	0.531	0.080
		(0.005)	(0.023)	(0.035)	(0.129)	(0.006)	(0.080)	(0.011)	(0.073)	(0.009)	(0.064)	(0.009)	(0.047)
	Qwen3-1.7B	0.812	0.474	0.716	0.330	0.801	0.473	0.789	0.404	0.700	0.320	0.525	0.051
		(0.004)	(0.056)	(0.008)	(0.043)	(0.006)	(0.058)	(0.002)	(0.043)	(0.004)	(0.023)	(0.011)	(0.057)
	Qwen3-4B	0.861	0.552	0.692	0.372	0.852	0.535	0.846	0.415	0.797	0.495	0.582	0.142
		(0.004)	(0.048)	(0.071)	(0.061)	(0.005)	(0.050)	(0.005)	(0.054)	(0.006)	(0.022)	(0.012)	(0.057)
	Qwen3-8B	0.884	0.579	0.711	0.376	0.878	0.570	0.878	0.506	0.814	0.433	0.669	0.217
Zh		(0.002)	(0.049)	(0.002)	(0.045)	(0.002)	(0.053)	(0.003)	(0.068)	(0.002)	(0.042)	(0.014)	(0.046)
	Llama-3.2-3B-Instruct	0.808	0.470	0.691	0.356	0.797	0.455	0.781	0.235	0.683	0.243	0.540	0.063
		(0.005)	(0.062)	(0.006)	(0.044)	(0.007)	(0.067)	(0.010)	(0.057)	(0.007)	(0.055)	(0.006)	(0.019)

Table 2: Performance on Empirical datasets. Best AUC (higher) and Spearman ρ (higher) per model are bolded.

AUC and report AUC on *test*; Spearman is computed on human dev sets.

Metric computation details. ROC-AUC is computed over the binary labels (is_positive) from the raw scores $NN_\theta(\phi(c, r))$ (no thresholding). Spearman’s ρ is computed between $NN_\theta(\phi(c, r))$ and the human relevance field annot_relevant_mean; ties are handled by the standard rank definition in scipy/sklearn. We report per-backbone results and also averages across backbones in the main figure.

D.2 Complete Results (Test AUC)

Table 2 reports ROC-AUC on the response-ranking task for all methods, split by language and backbone (*test only*).

D.3 Notes and Recommendations

Prompt choice matters. We use a single English prompt for both languages to keep training/evaluation invariant across cases; the Chinese template can be swapped in with no other changes. Pair-level prompting (single sequence for (c, r)) avoids the need for cross-encoder similarity and empirically stabilizes the scorer.

KDE caveat. Even with StandardScaler and PCA (128-d) plus an automatic bandwidth heuristic, KDE under-performs in high-dimensional spaces; results corroborate this limitation across both languages.

Replicability. All methods share the same negatives, scorer head, optimizer, and early-stopping rule; random seed is fixed (42). We rely on vLLM for embeddings (max_model_len=2048, GPU util 0.90) and compute AUC/Spearman with standard sklearn APIs.